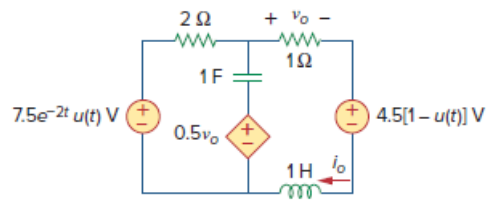


Problem

Find $i_o(t)$ for $t > 0$ in the circuit in Fig. 16.72.

Figure 16.72:



Step-by-step solution

Step 1 of 8

Refer to circuit diagram in Figure 16.72 in the textbook.

It is known that,

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

For $t < 0$, the unit step function is zero, the voltage source $7.5e^{-2t}u(t)$ V is replaced by the short circuit, that is $V = 0$ V and the voltage source $4.5(1-u(t))$ V is equal to 4.5 V.

Draw the circuit diagram for $t < 0$.

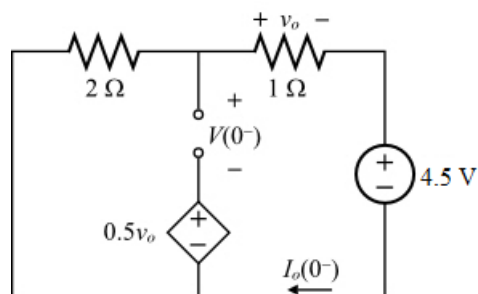


Figure 1

Step 2 of 8

Calculate the initial current through the inductor.

$$\begin{aligned} I_o(0^-) &= \frac{-4.5 \text{ V}}{1 \Omega + 2 \Omega} \\ &= \frac{-4.5 \text{ V}}{3 \Omega} \\ &= -1.5 \text{ A} \end{aligned}$$

Calculate the initial voltage across the capacitor.

$$\begin{aligned} V(0) &= -(2 \Omega) I_o(0^-) \\ &= -(2 \Omega)(-1.5 \text{ A}) \\ &= 3 \text{ V} \end{aligned}$$

Step 3 of 8

Draw the s-domain circuit diagram for $t > 0$.

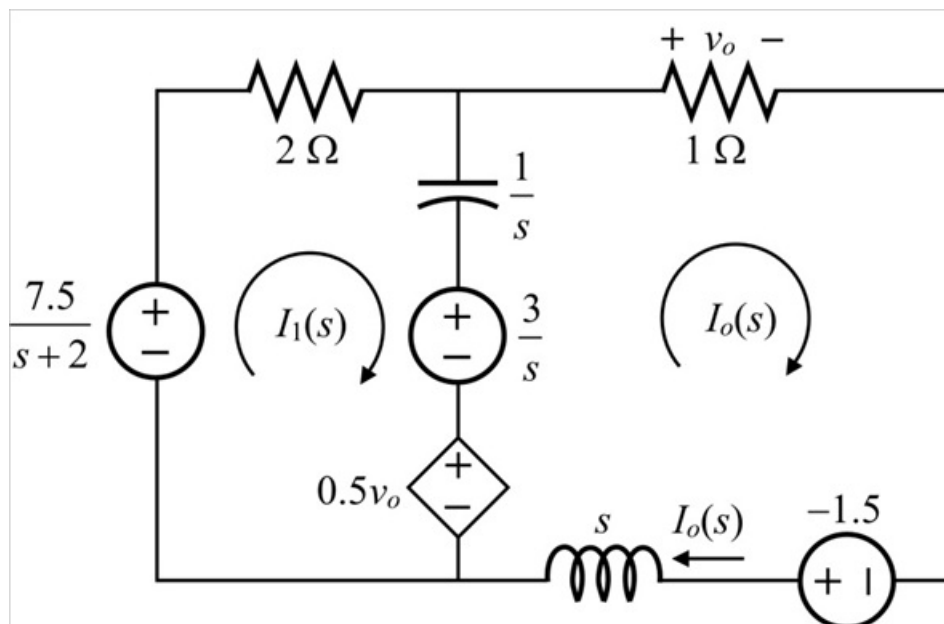


Figure 2

Step 4 of 8

From the circuit shown in Figure 2,

$$\begin{aligned} v_o &= (1 \Omega) I_o(s) \\ &= I_o(s) \end{aligned}$$

Apply Kirchhoff's voltage law to mesh $I_1(s)$.

$$-\frac{7.5}{s+2} + 2I_1(s) + \left(\frac{1}{s}\right)[I_1(s) - I_o(s)] + \frac{3}{s} + 0.5v_o = 0$$

Substitute $I_o(s)$ for v_o .

$$2I_1(s) + \left(\frac{1}{s}\right)[I_1(s) - I_o(s)] + 0.5I_o(s) = \frac{7.5}{s+2} - \frac{3}{s}$$

$$\left(2 + \frac{1}{s}\right)I_1(s) - \left(\frac{1}{s} - 0.5\right)I_o(s) = \frac{7.5}{s+2} - \frac{3}{s}$$

$$\left(\frac{2s+1}{s}\right)I_1(s) = \frac{7.5}{s+2} - \frac{3}{s} + \left(\frac{1-0.5s}{s}\right)I_o(s)$$

$$I_1(s) = \frac{7.5s}{(s+2)(2s+1)} - \frac{3}{2s+1} + \left(\frac{1-0.5s}{2s+1}\right)I_o(s) \dots\dots (1)$$

Step 5 of 8

Apply Kirchhoff's voltage law to mesh $I_o(s)$.

$$-0.5v_o - \frac{3}{s} + \left(\frac{1}{s}\right)[I_o(s) - I_1(s)] + (1)I_o(s) - (-1.5) + sI_o(s) = 0$$

Substitute $I_o(s)$ for v_o .

$$-0.5I_o(s) - \frac{3}{s} + \left(\frac{1}{s}\right)[I_o(s) - I_1(s)] + (1)I_o(s) - (-1.5) + sI_o(s) = 0$$

$$\left(-0.5 + \frac{1}{s} + 1 + s\right)I_o(s) - \frac{I_1(s)}{s} = \frac{3}{s} - 1.5$$

$$\frac{I_1(s)}{s} = \left(\frac{1 + 0.5s + s^2}{s}\right)I_o(s) - \frac{3}{s} + 1.5$$

$$I_1(s) = (1 + 0.5s + s^2)I_o(s) - 3 + 1.5s \dots\dots (2)$$

Step 6 of 8

Equate equation (1) and equation (2).

$$\frac{7.5s}{(s+2)(2s+1)} - \frac{3}{2s+1} + \left(\frac{1-0.5s}{2s+1}\right)I_o(s) = (1+0.5s+s^2)I_o(s) - 3 + 1.5s$$

$$\left(\frac{1-0.5s}{2s+1}\right)I_o(s) - (1+0.5s+s^2)I_o(s) = \frac{3}{2s+1} - \frac{7.5s}{(s+2)(2s+1)} - 3 + 1.5s$$

$$\left(\frac{1-0.5s - (1+0.5s+s^2)(2s+1)}{2s+1}\right)I_o(s) = \frac{3(s+2) - 7.5s - (3-1.5s)(s+2)(2s+1)}{(s+2)(2s+1)}$$

$$I_o(s) = \frac{3s+6-7.5s-6s^2-15s-6+3s^3+7.5s^2+3s}{(s+2)[1-0.5s-2s^3-1-2.5s-2s^2]}$$

$$= \frac{3s^3+1.5s^2-16.5s}{(s+2)[-2s^3-2s^2-3s]}$$

$$= \frac{s(3s^2+1.5s-16.5)}{-s(s+2)[2s^2+2s+3]}$$

$$= -\frac{3s^2+1.5s-16.5}{(s+2)(2s^2+2s+3)}$$

Step 7 of 8

Simplify the equation using partial fractions.

$$I_o(s) = \frac{A}{s+2} + \frac{Bs+C}{2s^2+2s+3} \dots\dots (3)$$

$$-\frac{3s^2+1.5s-16.5}{(s+2)(2s^2+2s+3)} = \frac{A(2s^2+2s+3) + Bs(s+2) + C(s+2)}{(s+2)(2s^2+2s+3)}$$

$$-3s^2 - 1.5s + 16.5 = A(2s^2+2s+3) + B(s^2+2s) + C(s+2) \dots\dots (4)$$

Equate constant terms from equation (4).

$$16.5 = 3A + 2C \dots\dots (5)$$

Equate s from equation (4).

$$-1.5 = 2A + 2B + C \dots\dots (6)$$

Equate s^2 from equation (4).

$$-3 = 2A + B \dots\dots (7)$$

Solve equations (5), (6) and (7).

$$A = 1.0714$$

$$B = -5.143$$

$$C = 6.643$$

Recall equation (3).

$$I_o(s) = \frac{A}{s+2} + \frac{Bs+C}{2s^2+2s+3}$$

Substitute 1.0714 for A , -5.143 for B and 6.643 for C .

$$\begin{aligned} I_o(s) &= \frac{1.0714}{s+2} + \frac{-5.143s+6.643}{2(s^2+s+1.5)} \\ &= \frac{1.0714}{s+2} + \frac{-2.572s+3.3215}{(s+0.5)^2 + (\sqrt{1.25})^2} \\ &= \frac{1.0714}{s+2} + \frac{-2.572(s+0.5)+4.6075}{(s+0.5)^2 + (\sqrt{1.25})^2} \\ &= \frac{1.0714}{s+2} - \frac{2.572(s+0.5)}{(s+0.5)^2 + (\sqrt{1.25})^2} + \frac{4.12\sqrt{1.25}}{(s+0.5)^2 + (\sqrt{1.25})^2} \end{aligned}$$

Apply inverse Laplace transform.

$$\begin{aligned} i_o(t) &= 1.0714e^{-2t}u(t) - 2.572e^{-0.5t} \cos(\sqrt{1.25}t)u(t) + 4.12e^{-0.5t} \sin(\sqrt{1.25}t)u(t) \\ &= \left[1.0714e^{-2t} + e^{-0.5t} \left\{ -2.572 \cos(\sqrt{1.25}t) + 4.121 \sin(\sqrt{1.25}t) \right\} \right] u(t) \text{ A} \end{aligned}$$

Therefore, the current, $i_o(t)$ for $t > 0$ is,

$$\boxed{\left[1.0714e^{-2t} + e^{-0.5t} \left\{ -2.572 \cos(\sqrt{1.25}t) + 4.121 \sin(\sqrt{1.25}t) \right\} \right] u(t) \text{ A} .}$$